

Network Administration and Management Lab. Work

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LIST OF CONTENTS

Homepage	1
1 Ubuntu Server Installation	2
1.1 Practical Objectives	2
1.2 Theoretical Background of Ubuntu Server	2
1.3 Ubuntu Server Practicum Steps	2
1.3.1 Installing Ubuntu Server in VirtualBox	3
1.3.2 Configuring the Network Adapter for External Access	3
1.4 Exercise	4
2 Samba Server	6
2.1 Practical Objectives	6
2.2 Theoretical Background of Samba Server	6
2.3 Samba Server Installation	7
2.4 1. Installation	7
2.5 2. Firewall Setup	8
2.6 Exercise	9
2.6.1 Setting Up File Sharing	9
2.6.2 Testing File Sharing from Windows	10
References	12

LIST OF FIGURES

1.1	Setting on VMWare via Adapter 2	3
1.2	Ubuntu Server	5
2.1	Illustration of Samba's ability to share resources across operating systems	6
2.2	Illustration of the client-server protocol	7
2.3	Samba running successfully	8
2.4	Samba inactive — installation may have failed	8
2.5	Samba shares visible in Windows File Explorer	11

LIST OF TABLES

1.2	Essential Linux Commands	5
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HOME PAGE

CHAPTER I UBUNTU SERVER INSTALLATION

1.1 Practical Objectives

- a. Students are able to install Ubuntu Server.
- b. Students understand and are able to practice basic commands in the Ubuntu Server environment.

1.2 Theoretical Background of Ubuntu Server

Ubuntu Server is a Linux-based server operating system based on the Ubuntu distribution, developed by Canonical Ltd. Ubuntu Server is specifically designed for use in server environments and network infrastructure. Ubuntu Server is the version of the Ubuntu operating system without a graphical user interface (headless), with a primary focus on performance and efficiency for server-related tasks [1].

The following are some important points about Ubuntu Server:

1. Open Source: Ubuntu Server is open-source, meaning that its source code can be freely accessed, used, modified, and distributed by anyone. As a result, many communities and organizations can contribute to the development and improvement of the operating system.
2. Stability and Security: Ubuntu Server is supported by a large and active community and is reviewed by security experts. Stable versions are released regularly with Long-Term Support (LTS), providing security updates and bug fixes for an extended period, typically up to five years after release.
3. APT Package Management: Like other Linux operating systems, Ubuntu Server uses the Advanced Package Tool (APT) as its package management system. This allows administrators to easily install, remove, and manage software and additional packages required by the server system.
4. Command-Line Interface (CLI): Ubuntu Server operates through a CLI, allowing administrators to manage the system remotely or through Secure Shell (SSH). The absence of a graphical interface helps reduce resource usage and improve security.
5. Cloud Support: Ubuntu Server is well known for its support for cloud computing environments. This server version is widely used by cloud service providers such as Amazon Web Services (AWS), Google Cloud Platform (GCP), and Microsoft Azure.
6. Virtualization Support: Ubuntu Server supports virtualization technologies such as Kernel-based Virtual Machine (KVM) and Docker containers. This allows users to create and manage virtual machines and containers easily and efficiently.
7. Access to Ubuntu Repositories: Ubuntu Server provides access to extensive Ubuntu repositories for downloading and installing various server software, including web applications, databases, email servers, and many others.
8. Documentation and Community Support: Ubuntu has a large and active community that can provide assistance through forums, discussion groups, and other online resources. In addition, paid support options are available from Canonical Ltd. for business and enterprise users.

Ubuntu Server has become a popular choice for building and managing server infrastructure due to its stability, security, and support for modern technologies such as cloud computing and virtualization. It provides powerful tools for operating servers reliably and efficiently [2]. The hands-on steps for installing and configuring it are covered in Section 1.3.

1.3 Ubuntu Server Practicum Steps

This guide walks through installing Ubuntu Server in VirtualBox and configuring its network adapter so the machine can be reached from outside VirtualBox.

1.3.1 Installing Ubuntu Server in VirtualBox

i What you'll need

An internet connection is required during installation, since packages are fetched while Ubuntu Server sets itself up.

1. **Download the Ubuntu Server installer image** from ubuntu.com/download/server. Choose the version that matches your computer's specifications. **Ubuntu Server 24** is the recommended version for this practicum.
2. **Install Ubuntu Server through VirtualBox.**

! Recommended virtual machine specifications

Setting	Recommended value
Base Memory	8 GB
Processors	2
Storage (VDI)	30 GB

Use a **different drive** from the one hosting your host operating system to store the VM folder!

1.3.2 Configuring the Network Adapter for External Access

The default network adapter in VirtualBox is isolated from the outside world. Adding a second, host-only adapter lets you reach the Ubuntu Server VM from your host machine, configured as shown in Figure 1.1.

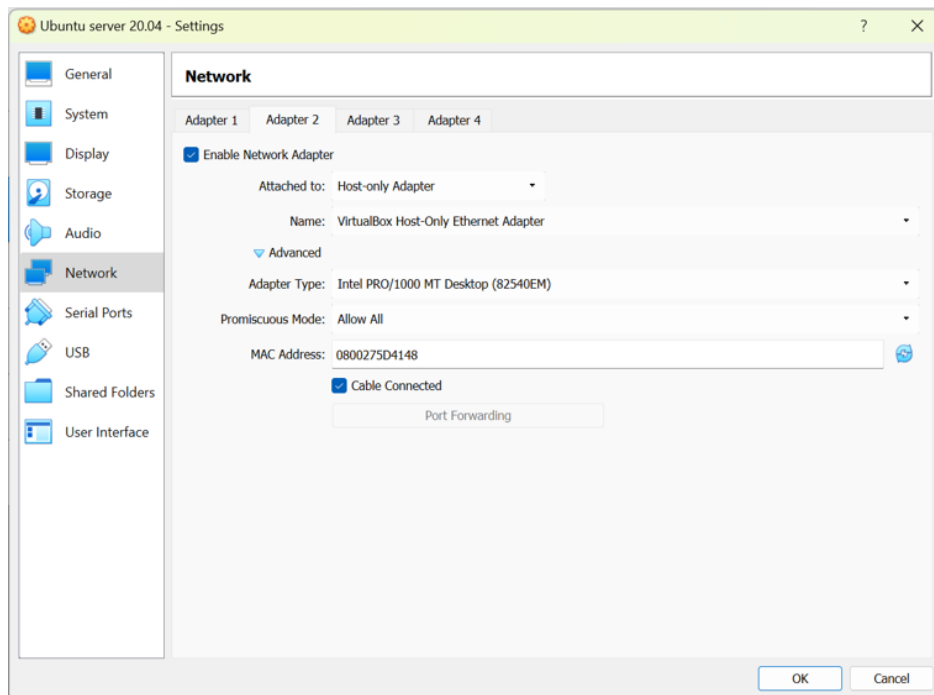


Figure 1.1: Setting on VMWare via Adapter 2

! Before you start

Power off the Ubuntu Server VM you just installed before changing its network settings.

1. **Enable a host-only adapter.**

In the VM's **Settings**, open the **Network** menu and select the **Adapter 2** tab. Then:

- Set **Attached to:** → Host-only Adapter
- Set **Promiscuous Mode:** → Allow All

2. **Power the VM back on** and edit the Netplan configuration [3].

Create a configuration file with:

```
sudo nano /etc/netplan/01-netconf.yaml
```

Add the following content to the file:

```
network:
  version: 2
  renderer: networkd
  ethernets:
    enp0s8:
      dhcp4: yes
```

Save and exit the editor by pressing Ctrl+X, then Y, then Enter.

3. **Apply the new network configuration.**

In the Ubuntu Server terminal, run:

```
sudo netplan apply
```

4. **Verify the IP address.**

Check that Ubuntu Server has received an IP address on the **enp0s8** adapter. You can confirm this with:

```
ip a
```

i Example result

In the example used for this guide, the Ubuntu Server IP address to be used in the next parts of the practicum is:

192.168.56.104

Figure 1.2 shows an example of this output in the Ubuntu Server terminal.

! Keep this address handy

The IP address obtained on the **enp0s8** adapter (e.g. 192.168.56.104) will be used to access this Ubuntu Server instance in later steps of the practicum. Write it down before continuing.

1.4 Exercise

Try the commands listed in Table 1.2, explain their functions, and provide screenshots of your results. Complete a total of 15 commands.


```

alam@alam:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3a:ea:f8 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86366sec preferred_lft 86366sec
    inet6 fe80::a00:27ff:fe3a:eaf8/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:55:f5:18 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.104/24 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 565sec preferred_lft 565sec
    inet6 fe80::a00:27ff:fe55:f518/64 scope link
        valid_lft forever preferred_lft forever

```

Figure 1.2: Ubuntu Server

Table 1.2: Essential Linux Commands

#	Command	#	Command	#	Command	#	Command
1	ls	22	ifconfig	43	lsof	64	parted
2	pwd	23	ip	44	dig	65	wc
3	cd	24	wget	45	nslookup	66	ls
4	clear	25	curl	46	du	67	nmap
5	mkdir	26	apt	47	tree	68	dmesg
6	mv	27	apt-get	48	ss	69	chattr
7	cp	28	yum	49	partx	70	usermod
8	rmdir	29	dnf	50	uptime	71	free
9	touch	30	rpm	51	tr	72	cron
10	cat	31	alias	52	ping	73	mysql
11	echo	32	dd	53	zcat	74	sdiff
12	less	33	top	54	xargs	75	history
13	tar	34	useradd	55	rm	76	netstat
14	grep	35	sleep	56	stat	77	sftp
15	head	36	screen	57	who	78	tcpdump
16	tail	37	pv	58	locate	79	scp
17	sort	38	fgrep	59	host	80	rsync
18	ps	39	dir	60	find	81	fsck
19	kill	40	egrep	61	fuser	82	bc
20	df	41	ssh	62	at	83	chage
21	chown	42	fd	63	fdisk	84	ffmpeg

CHAPTER II SAMBA SERVER

2.1 Practical Objectives

- a. Students are able to install Samba Server.
- b. Students are able to modify Samba configuration to manage user access rights and shared resources.
- c. Students are able to practice file sharing between Ubuntu and Windows using Samba.

2.2 Theoretical Background of Samba Server

Samba is a free and open-source software that provides an implementation of the SMB/CIFS protocol for non-Windows operating systems such as Linux, Unix, and macOS [4]. A Samba server allows these operating systems to function as a file and print server in a Windows network environment, enabling Windows users to share and access files and printers from computers running non-Windows operating systems, as illustrated in Figure 2.1. Samba provides features such as authentication, authorization, encryption, and access rights management to ensure the security and privacy of shared data. It can also be integrated with directory services such as Active Directory to simplify user management in large network environments [5].

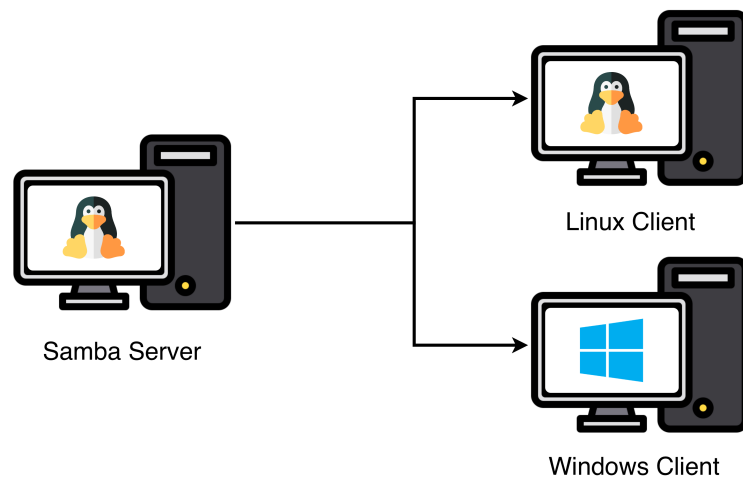


Figure 2.1: Illustration of Samba’s ability to share resources across operating systems

Samba operates using Server Message Block (SMB) protocol — a network protocol used to share files, printers, and other network resources between Windows or Unix computers [6]. SMB was originally developed by Microsoft as part of LAN Manager in the 1980s and has since become the de facto standard for file and printer sharing in Windows networks. SMB is a client-server protocol: the client sends requests to the server, and the server responds by delivering data or performing the requested operation, as illustrated in Figure 2.2. Data exchange is secured using encryption, authentication, and authorization.

Samba consists of two programs running in the background:

- **SMBD** — the file server; spawns a new process for each active client.
- **NMBD** — converts NetBIOS computer names to IP addresses and monitors shared resources on the network.

SMBD behavior is controlled through `/etc/samba/smb.conf` [7]. Samba can be configured as a

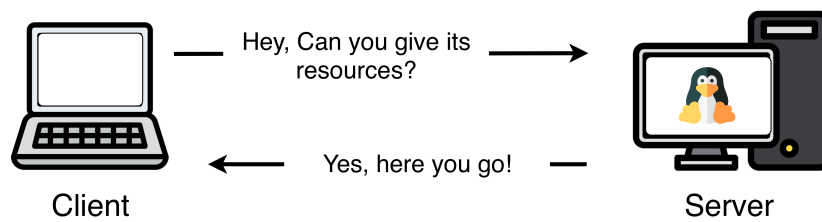


Figure 2.2: Illustration of the client-server protocol

file server, print server, or domain controller.

Uses of Samba Server

- Share one or more directory trees across the network.
- Centrally manage printers, print settings, and drivers for Windows clients.
- Assist clients with network browsing.
- Authenticate clients logging in to a Windows domain.

Advantages of Samba

- Free and open-source — no licensing cost.
- Available on multiple platforms for file sharing services.
- Easy to configure and administer.
- Connects directly to the network with minimal issues.
- Highly reliable — errors are uncommon unless the server has hardware problems.

Key Features of Samba

1. **User Authentication** — Users must log in before accessing Samba services. Samba accounts are managed with `smbpasswd`.
2. **Write Access Control** — Write permissions can be set per-share using `read only` or `writable` directives.
3. **Network Access Control** — Access can be restricted by allowing or blocking specific IP addresses.

2.3 Samba Server Installation

! Before you begin

These steps assume Ubuntu Server is already installed and running. If not, complete the Ubuntu Server installation on Chapter 1.

2.4 1. Installation

Step 1 — Log in as root

Log in to your Ubuntu Server terminal as root (or prefix each command with `sudo`).

Step 2 — Update the package list

```
apt update
```

Step 3 — Install Samba

```
apt install samba
```

①

- ① An active internet connection is required — packages are downloaded during installation.

Continue the installation until complete.

Step 4 — Verify the installation

```
systemctl status smbd
```

①

- ① **smbd** is the main Samba file-server daemon. A status of **active (running)** confirms a successful install.

If successful, the output will show **active (running)**, as shown in Figure 2.3.

```
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2022-08-30 01:37:05 UTC; 27s ago
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
  Process: 3485 ExecStartPre=/usr/share/samba/update-apparmor-samba-profile (code=exited, status=
 Main PID: 3504 (smbd)
   Status: "smbd: ready to serve connections..."
    Tasks: 4 (limit: 1066)
   Memory: 9.1M
    CGroup: /system.slice/smbd.service
            └─3504 /usr/sbin/smbd --foreground --no-process-group
              └─3506 /usr/sbin/smbd --foreground --no-process-group
                └─3507 /usr/sbin/smbd --foreground --no-process-group
                  └─3508 /usr/sbin/smbd --foreground --no-process-group
```

Figure 2.3: Samba running successfully

If the status shows **inactive (dead)** (Figure 2.4), try starting Samba manually:

```
service smbd start
```

If it still does not start, reinstall Samba.

```
● smbd.service - Samba SMB Daemon
   Loaded: loaded (/lib/systemd/system/smbd.service; enabled; vendor preset: enabled)
   Active: inactive (dead)
     Docs: man:smbd(8)
           man:samba(7)
           man:smb.conf(5)
```

Figure 2.4: Samba inactive — installation may have failed

2.5 2. Firewall Setup

Step 5 — Check firewall status

```
ufw status
```

If the firewall is not active, enable it first:

```
ufw enable
```

Allow Samba traffic through the firewall

```
ufw allow Samba
```

①

- ① This opens UDP ports 137–138 and TCP ports 139 and 445, which Samba needs to communicate with Windows clients.

 Do not skip this step

Without allowing Samba through the firewall, Windows clients will not be able to connect to the shared folders.

2.6 Exercise

2.6.1 Setting Up File Sharing

Step 1 — Create the shared directories

```
mkdir /home/share1  
mkdir /home/share2
```

①

- ① Both directories will be the root of each Samba share.

Create an empty test file inside each directory:

```
touch /home/share1/testfile1.txt  
touch /home/share2/testfile2.txt
```

Step 2 — Set directory permissions

```
chmod 777 /home/share1  
chmod 777 /home/share2
```

①

- ① 777 grants read, write, and execute permission to all users. For production use, more restrictive permissions are recommended.

Step 3 — Create Samba users

```
useradd user1  
useradd user2
```

①

- ① This creates a Linux system user. A separate Samba password is still required in the next step.

Step 4 — Set Samba passwords

```
smbpasswd -a user1
```

①

- ① The `-a` flag adds a new Samba user. Enter the password twice when prompted.

Repeat for `user2`:

```
smbpasswd -a user2
```

Step 5 — Configure access rights in `smb.conf`

Open the Samba configuration file:

```
nano /etc/samba/smb.conf
```

Scroll to the bottom and add the following blocks:

```
[share1]  
path = /home/share1 # <1>  
valid users = user1, user2 # <2>
```

```
read list = user2 # <3>
write list = user1 # <4>
browseable = yes

[share2]
path = /home/share2
valid users = user1, user2
read list = user1
write list = user2
browseable = yes
```

- ① The directory on the server that will be shared.
- ② Only these users may connect to this share.
- ③ **user2** can only read files in **share1**.
- ④ **user1** can read and write files in **share1**.

Save and exit by pressing Ctrl+X, then Y, then Enter.

Folder	user1	user2
share1	Read + Write	Read only
share2	Read only	Read + Write

Step 6 — Restart Samba

```
systemctl restart smbd
```

①

- ① Configuration changes only take effect after a restart.

2.6.2 Testing File Sharing from Windows

Step 1 — Find the Ubuntu Server IP address

```
ip a
```

①

- ① Look for the IP address on the **enp0s8** adapter (the host-only adapter configured earlier).

Note the IP address (e.g., 192.168.56.101).

Step 2 — Open the shares in File Explorer

On your Windows machine, open **File Explorer** and type the following in the address bar:

```
\\192.168.56.101
```

Confirm that both **share1** and **share2** are visible, as shown in Figure 2.5.

Step 3 — Test access as user1

Click **share1** and log in as **user1** when prompted. Check each of the following:

- Can you view the contents of **share1**?
- Can you create a new file inside **share1**?
- Can you copy a file from **share1** to local Windows storage?

Then navigate to **share2** (still logged in as **user1**) and repeat the same checks.

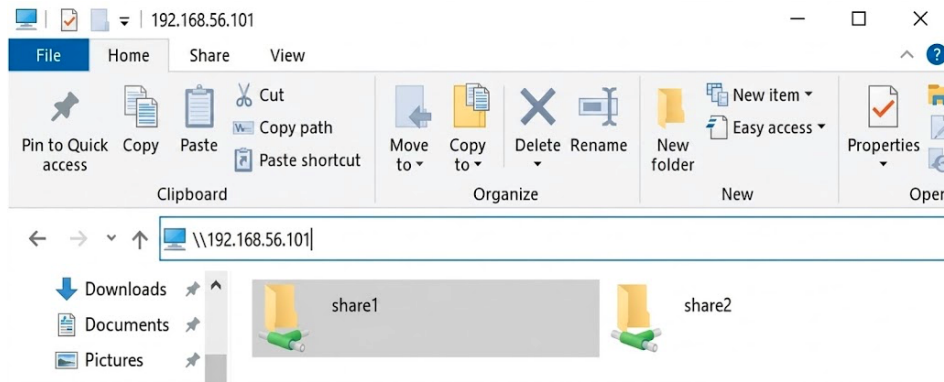


Figure 2.5: Samba shares visible in Windows File Explorer

💡 Expected result for user1

- **share1** → can view, create, and copy files
- **share2** → can view and copy files, but **cannot** create or modify files

Step 4 — Disconnect and switch to user2

Open **Command Prompt** on Windows and check the active connection:

```
net use
```

Disconnect it (replace \\name with the connection name shown above):

```
net use \\name /delete # <1>
```

- ① Disconnecting is necessary — Windows caches credentials, so the old session must be cleared before logging in as a different user.

Step 5 — Test access as user2

Connect again to \\192.168.56.101, log in as **user2**, and check both **share1** and **share2**.

💡 Expected result for user2

- **share1** → can view and copy files, but **cannot** create or modify files
- **share2** → can view, create, and copy files

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